

Member ID: _____

Time: _____

Rank: _____



DATABASE APPLICATIONS (240) **STATE 2023**

Production:

Job 1: Import Tables	_____ (100 points)
Job 2: Crosstab Query	_____ (100 points)
Job 3: Query	_____ (100 points)
Job 4: Report	_____ (100 points)
Job 5: Mailing Labels	_____ (50 points)
TOTAL POINTS	_____ (450 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

1. Ensure this test booklet contains Jobs 1-5.
2. Key all jobs according to the instructions given.
3. Correct any and all formatting, spelling or grammar errors. Use the formatting guide in the *Style & Reference Manual*.
4. Your name or initials should *not* appear on any work you submit. Use your Member ID in any occasion you would normally key your reference initials.
5. In the lower **right** hand corner of ALL work submitted (unless otherwise specified), key your Member ID and job number.
6. If you complete the event before the end of the time allotted, notify the proctor. Time may be considered a factor in determining a winner when there is a tie score.
7. Place your scoring sheet on top of your jobs. Jobs should be placed in numerical order.

	Points Possible	Points Earned
Job 1 – Import Tables		
Calculated field correct (Total Hours)	20	
All time fields formatted correctly (5 pts each)	20	
Table shows only info from 4/3/23	10	
Sort by Total Hours (smallest to largest)	10	
Hide fields: AutoNumber, AM Hours, PM Hours (10 pts each)	30	
Print in Landscape, using best fit, on one page	10	
Total for Job 1	100	
Job 2 – CrossTab Query		
Correct field names (10 pts each)	70	
Total weekly hours calculated for each employee	10	
Sort by Total Weekly Hours (largest to smallest)	10	
Print in Portrait, using best fit, on one page	10	
Total for Job 2	100	
Job 3 – Employee Hire Info Query		
Correct fields shown	20	
Calculate Avg Hourly Salary for each department	30	
Sort Descending by Average Salary	25	
Print in Portrait, using best fit, on one page	25	
Total for Job 3	100	
Job 4 – Weekly Salary Report		
Correct field names (all or nothing)	10	
Formatted as shown	5	
Group by Department	5	
Sorted by Last Name within each Department	5	
Department headings underlined & labeled	5	
Calculated field: Overtime Hours	10	
Calculated field: Weekly Pay	10	
Total 'Department' Hours label & Calculated field (Bold)	10	
Total Weekly Hours label & Calculated field (Bold)	10	
Line inserted under Weekly Hours and Weekly Pay sections	5	
Grand Total Hours: Label & Calculated field (bold)	10	
Grand Total Salary: Label & Calculated field (bold)	10	
Print in Portrait orientation, using best fit, & fit on one page	5	
Total for Job 4	100	
Job 5 – Mailing Labels		
Address labels formatted correctly	20	
Sort Ascending by Last Name	10	
Print with Member ID# and Job# in footer	10	
Print using best fit, on one page	10	
Total for Job 4	50	
Total Points Earned	450	

You've been hired by Tom Carlson, Information Technology manager, to create an employee database for Digital Solutions. Please follow his instructions for each of the jobs listed below.

Job 1 – Import Tables

1. Import the tables listed below into your database. Name the tables as shown below—without the “Student Data” portion.
 - a. Student Data Employee Hire Info (Primary Key = Employee ID)
 - b. Student Data Employee Info (Primary Key = Employee ID)
 - c. Student Data Time Clock (Primary Key = AutoNumber)
2. In the “Time Clock” table:
 - a. Create a “Total Hours” field that calculates the total hours each day for every employee
 - b. Change the format of all the time fields so that they show the time of day in hours and minutes (including AM/PM). For example, 6:45 AM
 - c. Filter the table to only show information from 4/3/23
 - d. Hide the following fields: “AutoNumber”, “AM Hours”, and “PM Hours”
 - e. Sort the table by “Total Hours” (smallest to largest)
 - f. Save the table as: “**Time Clock Table – XXXXX**” (where XXXXX is your Member ID#)
 - g. Print the table, in landscape orientation, make sure all data is visible and fits on one page

Job 2 – Create a Crosstab Query

1. Create a crosstab query that shows the daily hours for each employee and the total weekly hours for each employee.
2. Sort the query by the “Total Weekly Hours”—largest to smallest.
3. Save the Crosstab query as “**Crosstab Hours Query – XXXXX**” where ‘XXXXX’ is your Member ID.
4. Print the crosstab query, in portrait orientation, make sure all data is visible, and fits on one page.

Job 3 – Append Records to a Table and Create a Query

1. Append the “Employee New Hire” **DATA** to the “Employee Hire Info” table
2. Once you've appended the new hire data—create a query that calculates the average hourly salary for each department.
3. Sort the query, in descending order, by “Average Salary”.
4. Save the query as “**Employee Hire Info Query – XXXXX**” where ‘XXXXX’ is your Member ID#.
5. Print the query, in portrait orientation, make sure all data is visible, and fits on one page.

Job 4 - Weekly Salary Report

1. Create a report that is formatted like the one shown below:

Weekly Salary Report - XXXXX				
Administrative Support				
<u>Employee</u>	<u>Weekly Hours</u>	<u>Overtime Hours</u>	<u>Hourly Salary</u>	<u>Weekly Pay</u>
Adelsberger, Ryan	44		\$25.50	
Carsey, Haven	44.25		\$38.75	
Renick, Edna	41		\$49.75	
Total Administrative Support Department Hours:			Total Weekly Pay:	

2. The report should be grouped by “Department” and sorted by “Last Name” within each department.
3. The headings for each department should be underlined and labeled as shown above.
4. Create a calculated field that shows how many overtime hours each employee worked for the week. Anything over 40 hours is considered overtime.
5. Create a calculated field that shows the weekly pay for each employee. Employees are paid time-and-a-half for every overtime hour.
6. Create a label for the “Total Department Hours” that includes the name of each department.
7. Create a calculated field that sums the “Total Weekly Hours” for each department.
8. Create a label for “Total Weekly Pay” and create a calculated field that sums the “Total Weekly Pay” for each department.
9. The labels and calculated field in the “Department Total” section should be bold.
10. Insert a line under the “Weekly Hours” and the “Weekly Pay” sections—before the total for each department is selected.
11. At the bottom of the report, create a calculated field that sums **ALL** of the “Weekly Hours”. The label for this field should be “Grand Total Hours”. The label and the calculated field should be bold. The calculated field should be lined up so that it is underneath the “Weekly Hours” portion of the report.
12. At the bottom of the report, create a calculated field that sums **ALL** of the “Weekly Pay”. The label for this field should be “Grand Total Salary”. The label and the calculated field should be bold. The calculated field should be lined up so that it is underneath the “Weekly Pay” portion of the report.
13. Print the report, in portrait orientation, make sure all data is visible, and fits on one page.

Job 5 – Labels

1. Create a mailing label for all employees. You will use “Avery 8160” as the template. Sort the labels in ascending order by last name. You may need to adjust the margins so that all the data is shown on one page when printed.
2. Insert your Member ID# and Job 5 in the footer.
3. Print the labels.

Title First Name Last Name Street Address City, State Zip
